

Adaptive Exam Preparation System

Using Active Learning & Uncertainty Sampling

A smart quiz platform adapting to individual student levels in real-time.

Problem?

- Strong students get bored or underchallenged
- Weak students get discouraged quickly
- Grading takes unnecessary time

Does it matter?

For Professor

- Faster evaluation
- Personalized quiz paths
- Deeper understanding of student level

For Student

- Less early frustration
- Level matched questions
- Realistic evaluation

Solution

A platform that **adapts the quiz path** to the student.

How?

Estimate how a student will answer the *next* question, and choose the most informative one

And stop when we are sure about all his next answers

But **HOW?**

Let's start from the beginning

Data Collection

150

questions.json

```
{ question: 1+1?  
  options: {  
    "3": false  
    "2": true  
    "4": false }  
}
```

121

answers.csv

studentID, questionID, correct.

17,340

Feature Engineering

student

- # questions answered
- accuracy so far
- accuracy_easy so far
- accuracy_med so far
- accuracy_hard so far
- avg response time

sessionID

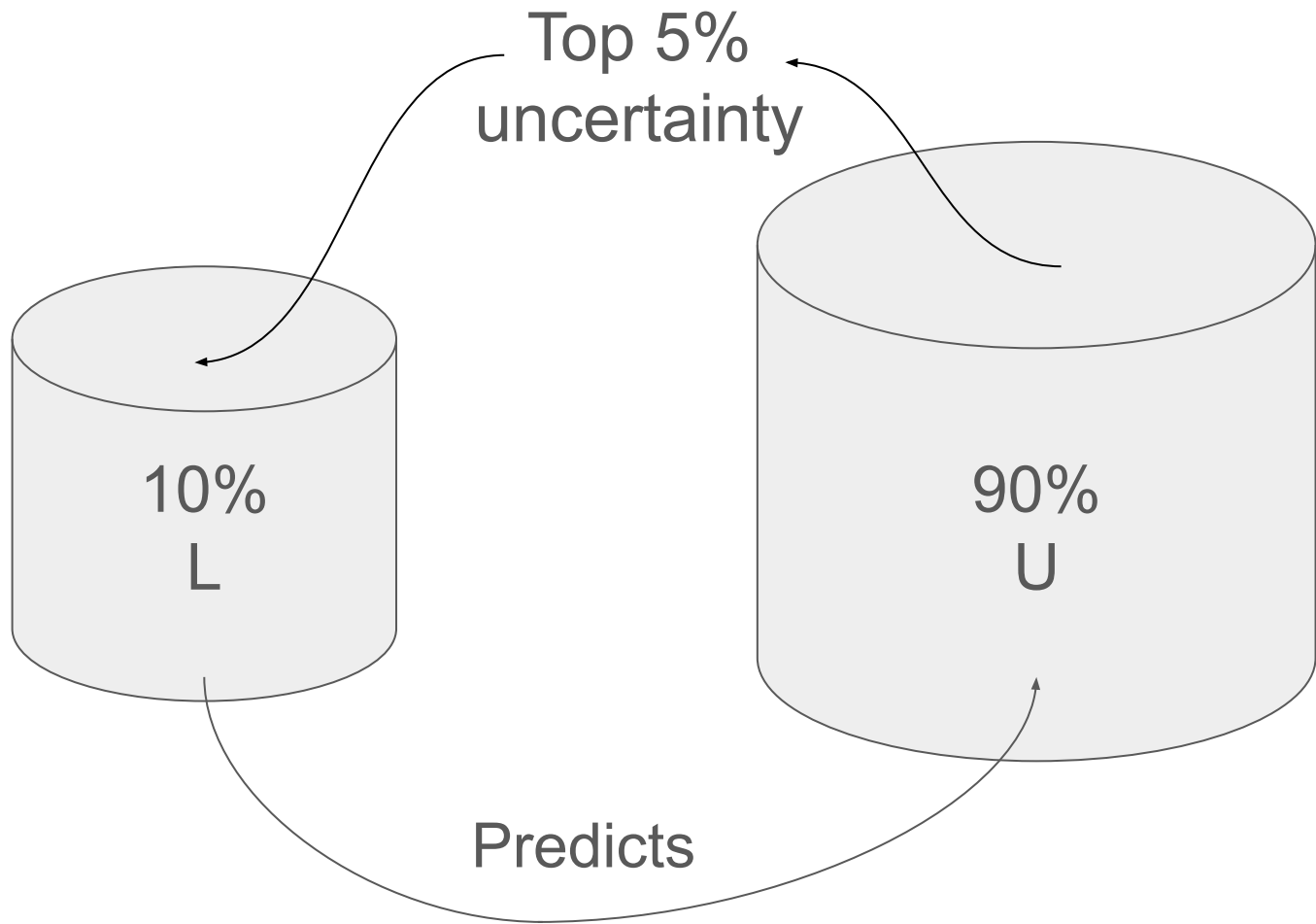
correct

Question

- difficulty
- complexity
- # of words
- # of options
- avg option length
- embedding

Now we **TALK**

Active learning **SIMULATION**



UNTIL?

Evaluation metric (LogLoss) stops improving

[xgboost]	Labeled: 1212 (10.00%)		LogLoss: 0.4053		AUC: 0.8335
[xgboost]	Labeled: 1757 (14.50%)		LogLoss: 0.3972		AUC: 0.8409
[xgboost]	Labeled: 2275 (18.77%)		LogLoss: 0.3926		AUC: 0.8436
[xgboost]	Labeled: 2767 (22.83%)		LogLoss: 0.3887		AUC: 0.8463
[xgboost]	Labeled: 3234 (26.68%)		LogLoss: 0.3868		AUC: 0.8451
[xgboost]	Labeled: 3678 (30.35%)		LogLoss: 0.3841		AUC: 0.8498
[xgboost]	Labeled: 4100 (33.83%)		LogLoss: 0.3818		AUC: 0.8509
[xgboost]	Labeled: 4501 (37.14%)		LogLoss: 0.3794		AUC: 0.8560
[xgboost]	Labeled: 4881 (40.27%)		LogLoss: 0.3777		AUC: 0.8568
[xgboost]	Labeled: 5242 (43.25%)		LogLoss: 0.3761		AUC: 0.8613
[xgboost]	Labeled: 5585 (46.08%)		LogLoss: 0.3725		AUC: 0.8648
[xgboost]	Labeled: 5911 (48.77%)		LogLoss: 0.3726		AUC: 0.8657

Model Selection

	model	best_logloss	final_logloss	final_labeled_ratio
0	xgboost	0.372532	0.372632	0.487706
1	random_forest	0.379131	0.379131	0.432508
4	gradient_boosting	0.380770	0.380770	0.402723
2	logistic_regression	0.401999	0.402973	0.303465
3	naive_bayes	0.421899	0.450861	0.460809

Insights

- Log Loss = 0.3726 → good probability quality, low prediction error, model is confident and mostly correct
- AUC = 0.8657 → strong ranking ability, model can clearly separate correct vs incorrect answers
- Brier Score = 0.1147 → good calibration, predicted probabilities are reasonably close to real outcomes
- 48% data used → huge training efficiency

 feature	# importance ...
difficulty	0.5486955
num_options	0.06468631
qstn_complexity	0.052699387
number_of_questions_answered_so_far	0.051657975
avg_option_length	0.051498115
num_words	0.04241999
accuracy_so_far	0.04054366
accuracy_easy_so_far	0.040499095
accuracy_medium_so_far	0.037144646
accuracy_hard_so_far	0.035234563

Simulation

BEGIN

State = initial state

Repeat {

Predict (correct | state , question features)

Choose difficulty based on accuracy so far

Choose most uncertain question within chosen difficulty

Student answers

Update state

} Until remaining predictions are <0.25 or >0.75

Score = $(\text{\#correct answers} + \sum \text{predictions}(\geq 0.5)) / \text{\# questions}$

END

Simulation

	session_id	# answered_count	# total_questions	# estimated_score	# real_score	# error	...
0	0Hpj01paAdfZ_q	17	30	0.4582098642985026	0.4666666666666666	0.0084568023681	
1	0Hpj01paAdfZ_q	5	30	0.6730203628540039	0.7	0.0269796371459	
2	0rzpTGhVxVJU_q	7	30	0.7243963877360026	0.7	0.0243963877360	
3	1ggBjCxnkaT8_q	17	30	0.4615735689798991	0.5	0.0384264310201	
4	1ggBjCxnkaT8_q	4	30	0.7010598500569661	0.8	0.0989401499430	
5	1ggBjCxnkaT8_q	7	30	0.7691385904947917	0.8	0.0308614095052	
6	1tnkUsBnGX2u_q	4	30	0.741029930114746	0.8	0.0589700698852	
7	1tnkUsBnGX2u_q	6	30	0.8093241373697917	0.7666666666666666	0.0426574707031	
8	2AQDG30NelzY_c	8	30	0.7046951293945313	0.8333333333333333	0.1286382039388	
9	5WCMaHHFn6W	9	30	0.7094404856363933	0.8	0.0905595143636	
10	5WCMaHHFn6W	4	30	0.7010598500569661	0.8	0.0989401499430	

Average error = 0.06 = 6%

Production

And the real **ACTIVE LEARNING**

Questions?